

Headphones are ubiquitous with music nowadays. Almost everyone – including audiophiles and people who want to enjoy their music and be left alone – owns a pair of earphones. Headphones are a broad category which encompass different styles of wear; on-ear, over the ear, in ear, earbuds and bone conduction. It is a common misconception that headphones evolved from speakers, but it is quite the contrary. Headphones were the only way to listen to audio signals before the invention of amplifiers which put the thump in the audio signals. According to the US Navy, the first pair of headphones were used by them and developed by Nathaniel Baldwin, who developed them in his kitchen!

Unlike what manufacturers and big brother would tell you, headphones don't work on magic, they are purely electronic and deal with bits and bytes. Audio signals are transmitted from a device such as a MP3 player (It's 2016!), Mobile Phones, computers and musical instruments. These signals are in digital format and in the format of 1s and 0s. The headphones, however are purely electrical and do not speak binary. That's where the DAC comes in. The DAC or the Digital Analog Converter (which we introduced in the first chapter) converts these 0s and 1s into waves which are then electrically transmitted to the headphone drivers, which are responsible for producing sound. So, the quality and the electronics of the headphones can only do so much, because the DAC is usually present on the device transmitting such as your phone. But that's changing now, with the infamous removal of the headphone jack from the iPhone 7 and the Moto Z. The headphone jack was the analog part of the DAC. With its removal, the iPhone 7 hands over the job of converting the digital signals to analog signals to the headphones themselves, through the lightning/USB-C port. The newer headphones which support these ports include a DAC in the headphones which converts. In this case, the DAC (and Amp, if needed) in the headphones need to be powerful enough to drive the complete power of the headphones. Here, the quality of the headphones is dictated by the DAC.

With the sheer number of headphone and earphone options available in the market today, it is difficult to pick out one which will suit you, but we have made it easier by categorising the various types of headphones.

Headphone Fit

The first is with the names themselves. Headphones typically refer to the style of wearing these earspeakers.